

The Binding of Isaac 3DS — Item Reference

Base Player Stats

Stat	Base Value	Formula / Notes
Damage	1.0	Min 0.5, applied as tear damage
Speed	2.2	Min 1.0, Max 4.0 (movement pixels/frame)
Fire Rate	0.0 (modifier)	Cooldown = $12 - \text{fire_rate} \times 2$ frames (min 3, max 30)
Range	120.0 px	Min 60, Max 300 (distance tear travels)
Max HP	6 (3 hearts)	Min 2, Max 16 (8 hearts). Each heart = 2 HP

Item Pool (20 Items)

Stat Boost Items

#	Item	Dam- age	Speed	Fire Rate	Range	HP	Special	De- scrip- tion
1	Penta- gram	+0.5	—	—	—	—	—	Pure damage boost
2	The Belt	—	+0.3	—	—	—	—	Pure speed boost
3	Wire Coat	—	—	+0.7	—	—	—	Signific- antly faster tears
4	Lunch	—	—	—	—	+2	—	Heals +2 HP on pickup, raises max HP
5	Spe- lunker Hat	—	—	—	+1.5	—	—	Large range in- crease (+45 px effect- ive)
6	Blood Martyr	+0.5	—	—	—	—	—	Pure damage boost (same as Pen- tagram)
7	Stig- mata	+0.3	—	—	—	+2	—	Dam- age + health hybrid
8	Sad Onion	—	—	+1.0	—	—	—	Largest single fire rate boost

#	Item	Dam- age	Speed	Fire Rate	Range	HP	Special	De- scrip- tion
9	Wire Hanger	—	—	+0.5	—	—	—	Moder- ate fire rate boost
10	Growth Hor- mones	+0.4	+0.2	—	—	—	—	Dam- age + speed hybrid
11	Jesus Juice	+0.5	—	—	+0.5	—	—	Dam- age + range hybrid (+15 px effect- ive)

Multi-Stat Items

#	Item	Dam- age	Speed	Fire Rate	Range	HP	Special	De- scrip- tion
12	Speed Ball	—	+0.3	+0.5	—	—	—	Speed + fire rate combo
13	Magic Mushroom	+0.5	—	—	+1.0	+2	—	Dam- age + range (+30 px) + health. One of the best items
14	The Halo	+0.3	+0.2	+0.2	+0.5	+2	—	Boosts ALL five stats. Best all- rounder

Power Items (with Tradeoffs)

#	Item	Dam- age	Speed	Fire Rate	Range	HP	Special	De- scrip- tion
15	Poly-phemu s	+1.5	—	−1.0	—	—	—	Massive dam- age, much slower tears
16	Sacred Heart	+1.0	−0.3	—	—	—	 Homing	Huge damage + hom- ing, re- duced speed

Special Ability Items

#	Item	Dam- age	Speed	Fire Rate	Range	HP	Special	De- scrip- tion
17	Cu- pid's Arrow	—	—	—	—	—	 Pier- cing	Tears pass through en- emies
18	Spoon Bender	—	—	—	—	—	 Homing	Tears track nearest enemy
19	Inner Eye	—	—	-1.5	—	—	 Triple Shot	Fires 3 tears, much slower fire rate
20	Spirit Sword	—	—	—	—	—	 Spectral	Tears pass through walls/ obstacl es

Special Ability Flags

Flag	Bit	Effect
ITEM_FLAG_HOMING	1 << 0	Tears steer toward nearest enemy (0.18 strength/frame)
ITEM_FLAG_PIERCING	1 << 1	Tears don't disappear on enemy hit
ITEM_FLAG_SPECTRAL	1 << 2	Tears pass through walls and obstacles
ITEM_FLAG_TRIPLE	1 << 3	Fires 3 tears (center + 2 angled at ± 0.25 spread)

Flags are **cumulative** — collecting both Cupid's Arrow and Spoon Bender gives homing + piercing tears.

Stat Application Details

How Stats Stack

All item stats are **additive**. When a player collects an item:

1. All held items' bonuses are re-summed from base stats
2. Stats are clamped to min/max bounds
3. Ability flags are OR'd together

Fire Rate Formula

```
tear_cooldown = max(3, min(30, 12 - fire_rate_modifier × 2))
```

- Base: 12 frames between shots (~5 tears/sec at 60fps)
- With Sad Onion (+1.0): cooldown = 10 frames (~6 tears/sec)
- With Wire Coat + Sad Onion (+1.7): cooldown = 8.6 → 8 frames (~7.5 tears/sec)
- Minimum possible: 3 frames (~20 tears/sec)

Range Formula

```
effective_range = base_range + range_bonus × 30.0
```

Range bonus is multiplied by 30 before adding to the 120px base.

HP on Pickup

Items with HP bonus heal the player for that amount on pickup AND increase max HP permanently.

Item Spawn System

- Items appear on **pedestals** in **Treasure Rooms** (one per dungeon floor)
- Each treasure room contains exactly one random item
- Items are drawn from the full pool of 20 items randomly
- Maximum 16 items can be held simultaneously
- Item sprites are rendered from `ui_items.t3x` atlas on pedestals

Tear Visual Variants

Items affect tear appearance via the bullet atlas system:

Condition	Tear Sprite	Color	Effect
Normal	Blue tear (7 sizes)	Blue	Standard blue tear
Homing	Dark blue tear	Purple tinted	Rotation follows travel direction
Spectral	Blue tear	Ghostly/transparent	Semi-transparent rendering
Piercing	Any + trail	Same + fading copies	2 trailing ghost copies behind
High damage (>2.0)	Larger variant	Same	1.2× scale multiplier
Triple Shot	3× normal	Same	Center + 2 angled projectiles

Item Tier List (Subjective Power Ranking)

S Tier — Game-Changing

Item	Why
Sacred Heart	+1.0 dmg AND homing. Best offensive item in the game
Magic Mushroom	+0.5 dmg, +30 range, +2 HP. Incredible value
The Halo	Boosts every stat including +2 HP. No downsides

A Tier — Excellent

Item	Why
Polyphemus	+1.5 dmg is massive, fire rate penalty is manageable
Inner Eye	Triple shot triples effective DPS despite slower fire rate
Cupid's Arrow	Piercing is extremely strong in rooms with lined-up enemies
Spoon Bender	Homing eliminates need to aim, great for mobile enemies

B Tier — Good

Item	Why
Sad Onion	+1.0 fire rate is very noticeable
Spirit Sword	Spectral tears bypass cover, strong in obstacle-heavy rooms
Speed Ball	Speed + fire rate combo, no downsides
Jesus Juice	Solid dmg + range combo

C Tier — Decent

Item	Why
Pentagram	+0.5 dmg is always welcome
Blood Martyr	Same as Pentagram
Wire Coat	Good fire rate boost
Growth Hormones	Decent dmg + speed
Stigmata	Damage + healing is nice
Spelunker Hat	Range is situationally useful

D Tier — Filler

Item	Why
The Belt	Speed-only, less impactful
Wire Hanger	Smaller fire rate boost
Lunch	HP-only, no offensive benefit